

Experiment	<i>Lab 01: Experiment INT – Wind Tunnel Characterization</i>
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Section Number	04 – Robert Bryan

Summary

The goal of this lab was to learn how to set the airspeed in the test section and run the wind tunnel. In a brief introduction, the wind tunnel's various parts were described and displayed. Next, two experiments using a traverse pitot probe were conducted: one in which the fan's frequency changed while the probe remained stationary, and another in which the fan's frequency remained constant, but the probe's height changed. Through these trials, it was observed that speed increases with frequency, whereas speed declines slightly and air turbulence increases as height varies.

Aim & Objectives

The aim of this lab is to learn about the wind tunnel and how to gather and analyze data from it.

The objectives were:

- Operating the wind tunnel
- Learn about the different components of the wind tunnel system
- Collecting data using MATLAB and analyzing it
- Calculating all the uncertainties

Uncertainty Analysis

Airspeed (m/s)	Velocity Uncertainty (%)
2.89	0.4217
6.57	0.1583
10.4	0.0975
14.3	0.0730
18.2	0.0562

Table 1: Uncertainty data for velocity

Frequency (Hz)	Pressure Uncertainty (%)
5	1.2734
10	1.2567
15	1.2297
20	1.2661
25	1.2429

Table 2: Uncertainty data for variable pressure

The values were calculated using MATLAB code which can be found in the appendix.

Results & Discussion

This lab consisted of three sections. The first section introduced the wind tunnel and its components. The second section examined the relationship between airspeed and fan frequency while keeping the traverse Pitot tube at a fixed height. The third section analyzed airspeed by varying the height of the traverse Pitot tube while maintaining a constant airspeed.

1. Wind Tunnel Components:



Figure 1.1: Full Wind Tunnel

The figure shows the full wind tunnel consisting of the inlet, nozzle, test section, diffuser, and fan:

- a) **Inlet:** The air is sucked in through the inlet. It also consists of a mesh on the inside that stabilizes the flow and ensures no debris flows inside the wind tunnel.
- b) **Nozzle:** Used to accelerate the air flowing into the test section
- c) **Test Section:** Place where the objects are tested. It consists of pitot tubes, sensors, and data collectors connected to the data acquisition computer.
- d) **Diffuser:** Slows down the airflow and increases pressure.
- e) **Fan:** Sucks and allows air into the wind tunnel. Its speed is operated by the controller outside the test section which changes the frequency of the fan (Hz).

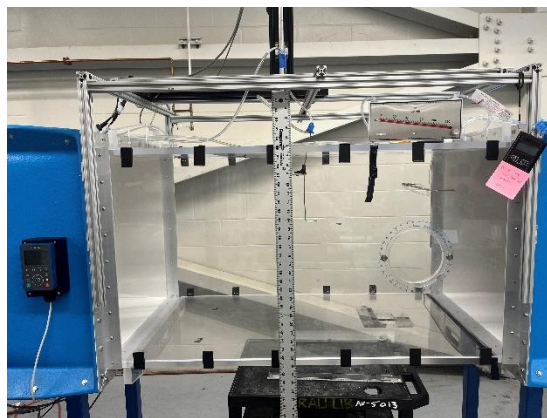


Figure 1.2: Test Section

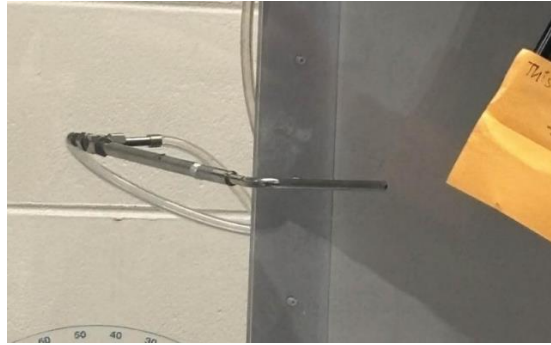


Figure 1.3: Pitot-Static Probe in the Test Section



Figure 1.4: Analog Manometer connected to Pitot-Static Tube



Figure 1.5: Analog-to-Digital Converter (ADC)



Figure 1.6: Data-Acquisition Computer

2. **Airspeed: Height is Constant, Speed Varies**

We start the experiment by measuring the ambient temperature (21.5 °C) and pressure (30.3 *inHg*). These were used to determine the density of air, shown in Appendix A. The velocity at various airspeeds can be calculated using the air density. In this part of the experiment, we measured the pressure at a fixed height while varying the fan frequency. We started with a baseline frequency of 0 Hz and raised it in increments of 5 Hz from 0 Hz to 25 Hz. The 0 Hz measurement was subtracted from the other measurements to determine the differential pressure. The airspeed was computed from this, assuming that higher frequencies would translate into faster speeds. Speed is calculated using:

$$V = \sqrt{\frac{2q}{\rho_{air}}}$$

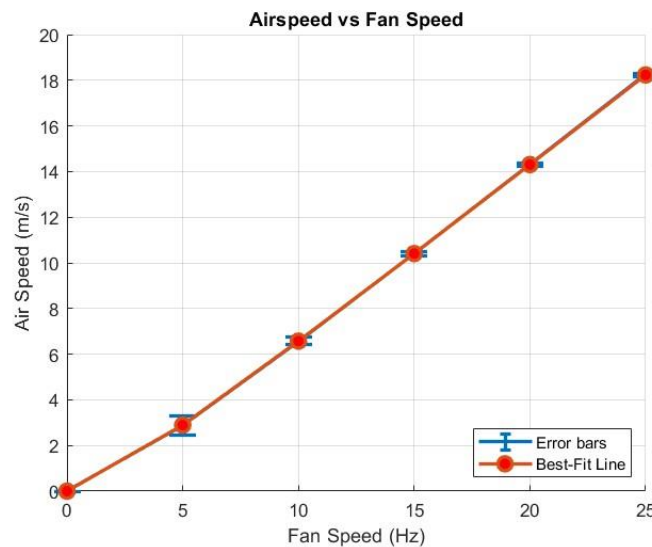


Figure 2: Airspeed vs Fan speed plot

The results validated our expectations, demonstrating that airspeed increases as the fan frequency rises, indicating a direct correlation. The uncertainty in airspeed measurements is summarized in Table 3.

Next, we can determine the Reynolds number per unit length using the formula provided below.

$$\frac{Re}{l} = \frac{\rho v}{\mu}$$

Appendix B shows the formula for μ .

Fan speed (Hz)	Airspeed (m/s)	Uncertainty	$\frac{Re}{l} \left(\frac{1}{m}\right)$
5	2.89	0.4217	192508
10	6.57	0.1583	437639
15	10.4	0.0975	692762
20	14.3	0.0730	952548
25	18.2	0.0562	1212334

Table 3: Reynolds number per unit length for each fan frequency

3. Airspeed: Height Varies, Speed is Constant

In this section of the experiment, we monitored the pressure at different heights ranging from 1 to 40 inches while maintaining a constant frequency of 15 Hz to maintain a velocity of 10 m/s. As the probe height increased, we assumed that the velocity would decrease.

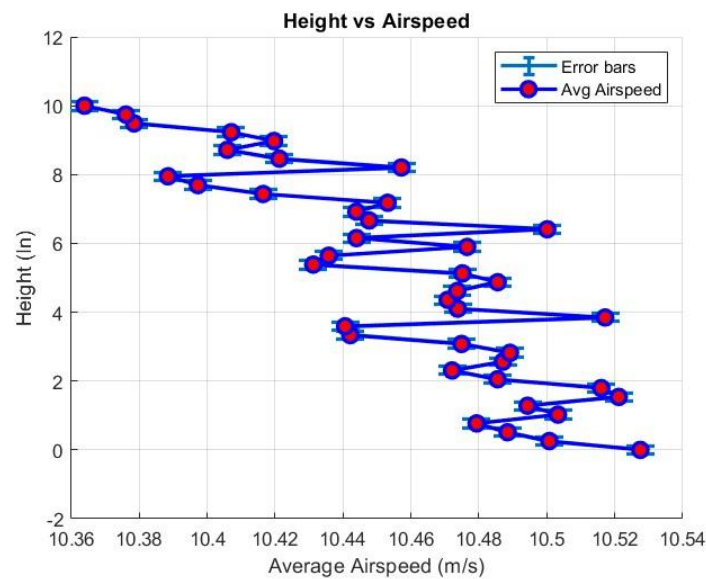


Figure 3: Height vs Airspeed

Due to the slight drop in pressure that occurs as height increases, it was found that the speed dropped as height increased approaching the wind tunnel's ceiling.

We can now calculate the Turbulence intensity using the following formula:

$$I = \frac{STD_v}{\bar{V}} * 100$$

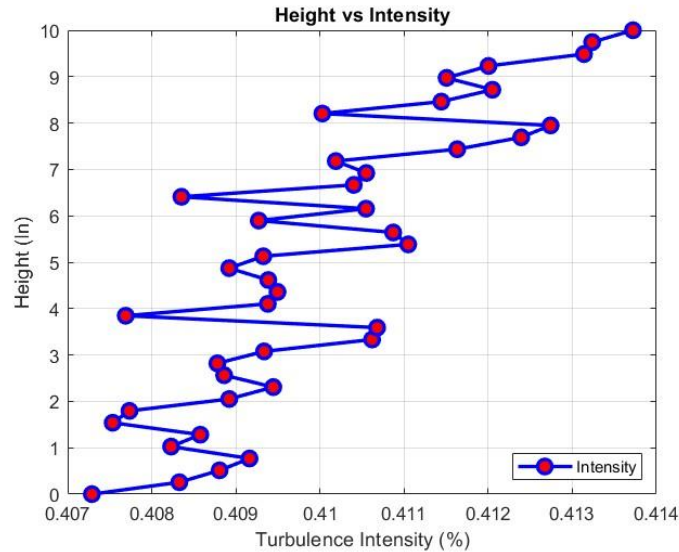


Figure 4: Height vs Turbulence Intensity

From the above plot, it resembles a mirrored version of Fig 3. Turbulence intensity is a dimensionless quantity that indicates the strength of turbulence within the fluid, which in this case refers to the turbulence in the air. We can infer that as the probe moved upward, it induced greater turbulence in the air, leading to a reduction in speed.

References

1. Experiment INT – Wind Tunnel Characterization: (2025). Embry-Riddle Aeronautical University.

Appendix

Appendix A: Density Calculation

$$P = \rho RT$$

$$\rho_{air} = \frac{P}{RT}$$

$$\rho_{air} = \frac{102607.6}{287 * 294.65} = 1.213 \frac{kg}{m^3}$$

Appendix B: Sutherlands Formula

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{\frac{3}{2}} * \frac{T_0 + S_\mu}{T + S_\mu}$$

$$(for\ air)\mu_0 = 1.716 * 10^{-5} \frac{N * s}{m^2}, T_0 = 273\ K, S_\mu = 111\ K$$

$$\mu = 1.716 * 10^{-5} \left(\frac{294.65}{273} \right)^{\frac{3}{2}} * \frac{273 + 111}{294.65 + 111} = 1.821 * 10^{-5} \frac{N * s}{m^2}$$

Appendix C: MATLAB Code

```
% Author: Anuranan Bharadwaj
% Lab 01: Experiment INT
```

```
clc;
clear;
close all;
```

```
data = readmatrix('SinglePoint_20250131T084139.csv'); %loading data from excel file
```

```
% Minus data and coverting inH2O to Pa
```

```
Hz_5 = (data(:,2) - data(:,1)) * 248.84;
Hz_10 = (data(:,3) - data(:,1)) * 248.84;
Hz_15 = (data(:,4) - data(:,1)) * 248.84;
Hz_20 = (data(:,5) - data(:,1)) * 248.84;
Hz_25 = (data(:,6) - data(:,1)) * 248.84;
```

```
% Equation of State: Patm = rho * R * Tatm
rho = 1.213; % kg/m^3
```

```
% calculating V: V=sqrt(2q/rho)
```

```
V_5 = sqrt(2*Hz_5/rho);
V_10 = sqrt(2*Hz_10/rho);
V_15 = sqrt(2*Hz_15/rho);
V_20 = sqrt(2*Hz_20/rho);
V_25 = sqrt(2*Hz_25/rho);
```

```
% Plot 1
```

```
fan_speeds = [0, 5, 10, 15, 20, 25]; % x-axis
air_speeds = [0, mean(V_5), mean(V_10), mean(V_15), mean(V_20), mean(V_25)]; % y-axis
```

```
%Error Bar
```

```
error_bar = [0, std(V_5), std(V_10), std(V_15), std(V_20), std(V_25)];
```

```
% Create figure
```

```
figure;
hold on;
```

```
errorbar(fan_speeds, air_speeds, error_bar, 'LineWidth', 2, 'CapSize', 15); % uncertainty
```

```
plot(fan_speeds, air_speeds, '-o', 'LineWidth', 2, 'MarkerSize', 8, 'MarkerFaceColor', 'r');
```

```
xticks(fan_speeds);
xticklabels({'0', '5', '10', '15', '20', '25'});
xlabel('Fan Speed (Hz)');
ylabel('Air Speed (m/s)');
title('Airspeed vs Fan Speed');
legend({'Error bars', 'Best-Fit Line'}, 'Location','southeast');
```



```
grid on;
hold off;
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Second Half %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
%data loading
```

```
data_Hz_0 = readmatrix('0Hz_Traverse_20250131T085646.csv');
data_Hz_15 = readmatrix('15Hz_Traverse_20250131T090457.csv');
```

```
%difference
```

```
data_HZ_diff = (data_Hz_15 - data_Hz_0) * 248.84; %coverting inH2O to Pa
```

```
%speed
```

```
v = sqrt((2*data_HZ_diff)/rho);
height = linspace(0, 10, size(data_HZ_diff, 1));
```

```
%average speed
```

```
v_2 = mean(v, 2); %computing mean of all columns giving (40x1)
```

```
%error bar
```

```
errorBar = std(v, 0 ,2);
```

```
%%%%%%%% Height vs Speed %%%%%%%%%
```

```
figure;
hold on;
```

```
errorbar(v_2, height, errorBar, 'LineWidth', 2, 'CapSize', 15); %uncertainty
plot(v_2, height, '-o', 'Color', 'b', 'LineWidth', 2, 'MarkerSize', 8, 'MarkerFaceColor', 'r')
```

```
title('Height vs Airspeed')
xlabel('Average Airspeed (m/s)');
ylabel('Height (In)');
legend({'Error bars', 'Avg Airspeed'});
```

```
grid on;
hold off;
```

```
%%%%%%%% Height vs Intensity %%%%%%%%%
```

```
intensity = (std(v_2) ./ v_2)*100;
```

```
figure;
```

```
plot(intensity, height, '-o', 'Color', 'b', 'LineWidth', 2, 'MarkerSize', 8, 'MarkerFaceColor', 'r');
title('Height vs Intensity');
```

```
xlabel('Turbulence Intensity (%)');  
ylabel('Height (In)');  
legend({'Intensity'}, 'Location', 'southeast');
```

```
grid on;
```

```
%%%%%%%% Uncertainty %%%%%%%%%%
```

```
%Velocity
```

```
UC_V_5 = (std(V_5) / sqrt(length(V_5))) * 100;  
UC_V_10 = (std(V_10) / sqrt(length(V_10))) * 100;  
UC_V_15 = (std(V_15) / sqrt(length(V_15))) * 100;  
UC_V_20 = (std(V_20) / sqrt(length(V_20))) * 100;  
UC_V_25 = (std(V_25) / sqrt(length(V_25))) * 100;
```

```
%Variable Pressure
```

```
UC_Hz_5 = (std(Hz_5) / sqrt(length(Hz_5))) * 100;  
UC_Hz_10 = (std(Hz_10) / sqrt(length(Hz_10))) * 100;  
UC_Hz_15 = (std(Hz_15) / sqrt(length(Hz_15))) * 100;  
UC_Hz_20 = (std(Hz_20) / sqrt(length(Hz_20))) * 100;  
UC_Hz_25 = (std(Hz_25) / sqrt(length(Hz_25))) * 100;
```